Reinforcement Learning - Sutton and Barto

Book Club Highlights

Alexandra Ortan

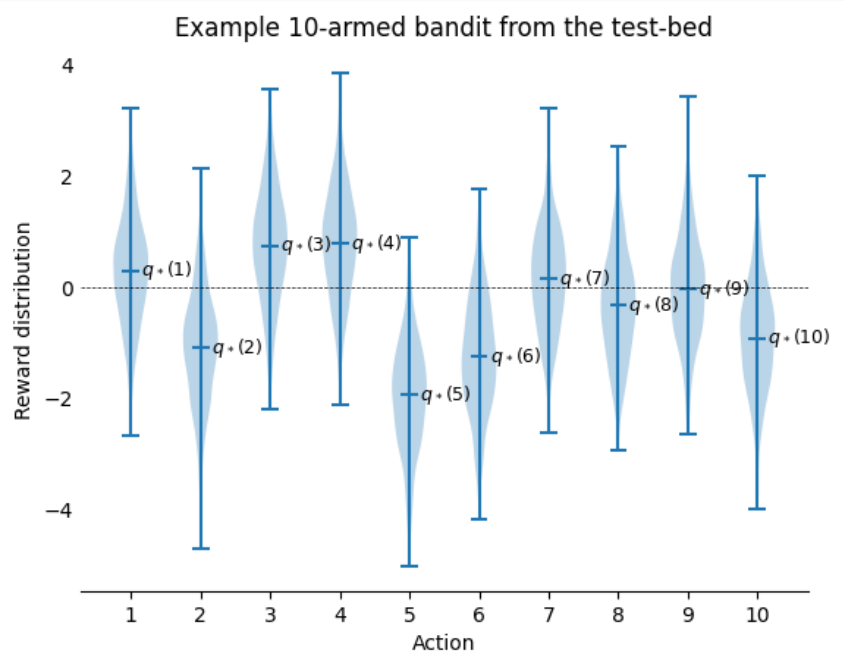
November 10, 2026

Chapter 2
Multi-armed Bandits

The 10-armed Testbed

Experimental setup

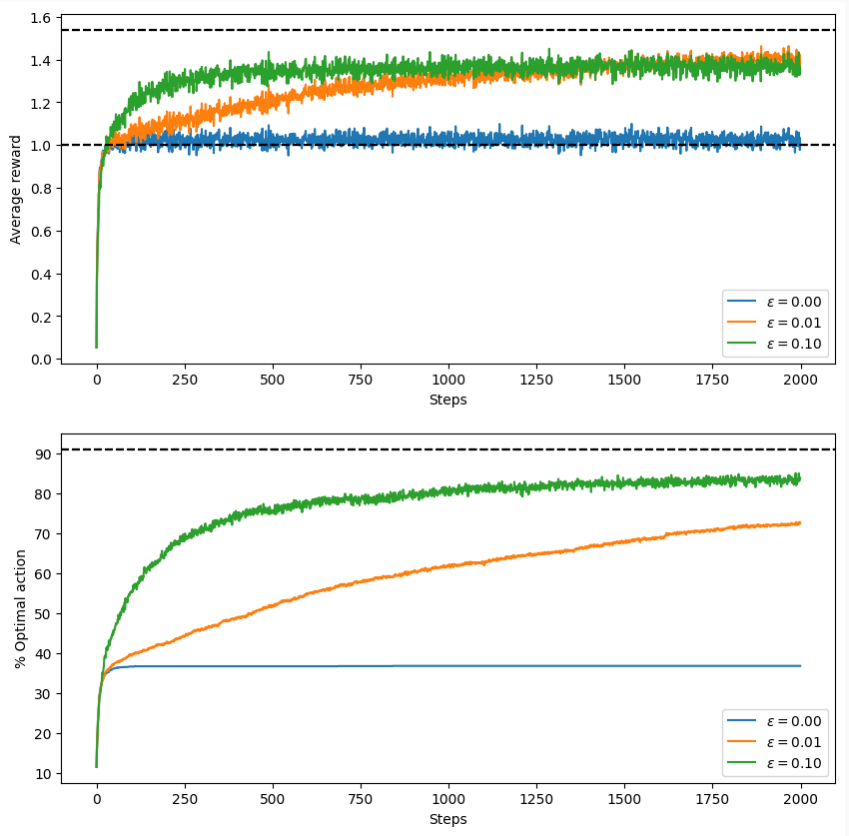
- $\rightarrow$ 2000 randomly generated 10-arm bandits
- $\rightarrow$ $q_*(a) \sim \mathcal{N}(0, 1)$
- $\rightarrow$ $R(a) \sim (q_*(a), 1)$



ϵ-greedy agent

Experiment

- $\rightarrow$ ϵ-greedy agent
- $\rightarrow$ sample average value estimator
- $\rightarrow$ reward at each step averaged over 2000 bandits



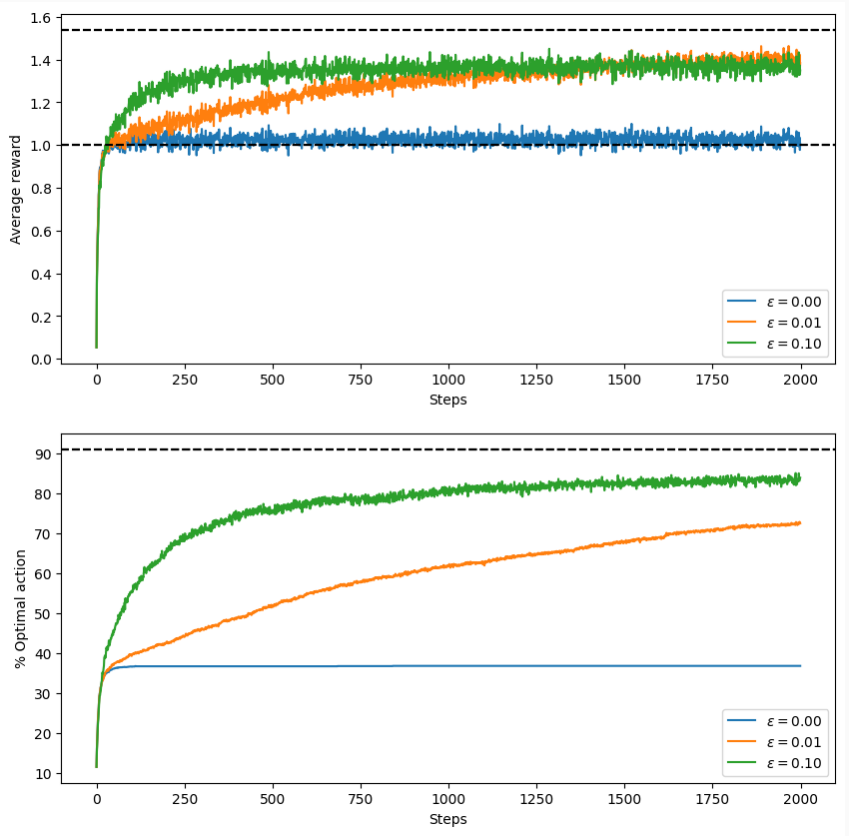
ϵ-greedy agent

Observation 1:

For $\epsilon = 0$, average reward ≈ 1

What gives?

- $\rightarrow$ This is the expected value of the first positive reward received.
- $\rightarrow$ Can be estimated via integrals weighted by survival functions - the chance that a random walk stays positive forever.
- $\rightarrow$ Depends on initial action value estimates (0 here).
- $\rightarrow$ Value is linear in the variance of the expected reward over actions.



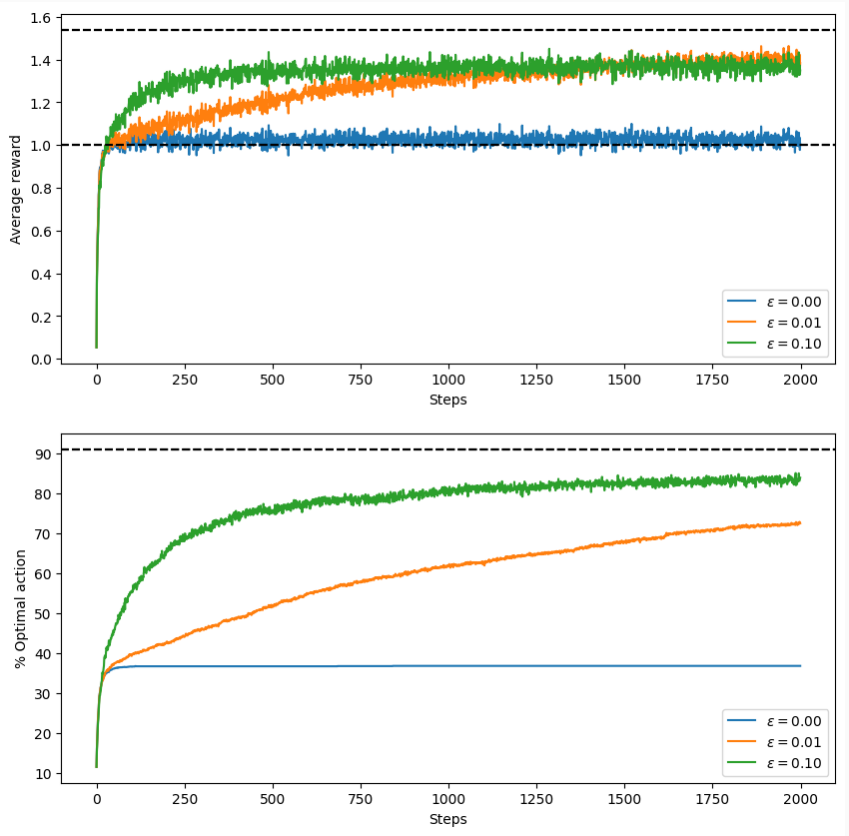
ϵ-greedy agent

Observation 2:

Average reward $\lesssim 1.54$

What gives?

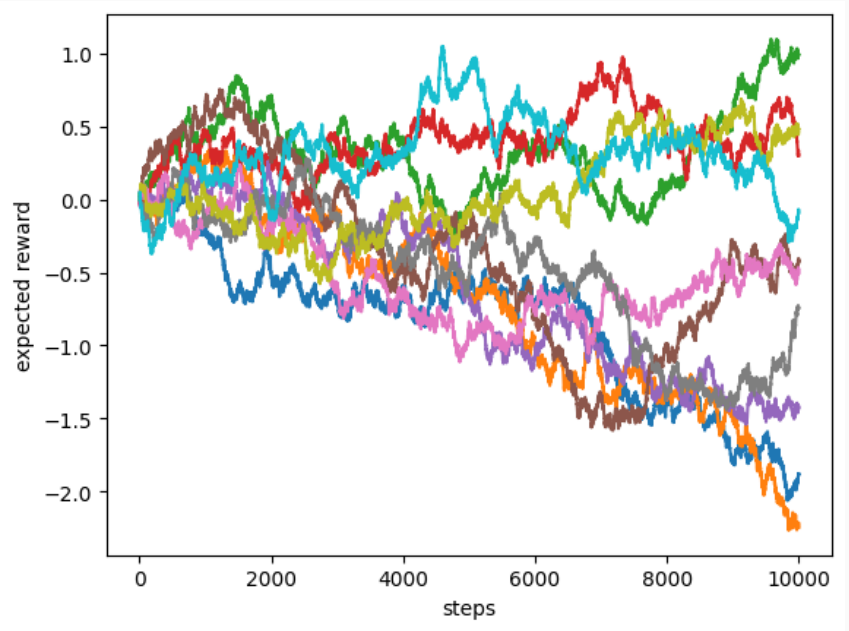
- $\rightarrow$ Theoretical ceiling, as provided by an oracle agent.
- $\rightarrow$ This is the best **any** agent can do.
- $\rightarrow$ Calculated as the expected value of the highest of k action values.
- $\rightarrow$ Ceiling $\approx \sqrt{2 \ln k}$
- $\rightarrow$ For $k = 10$, this ceiling ≈ 1.539.



Non-Stationary 10-armed Testbed

Experimental setup

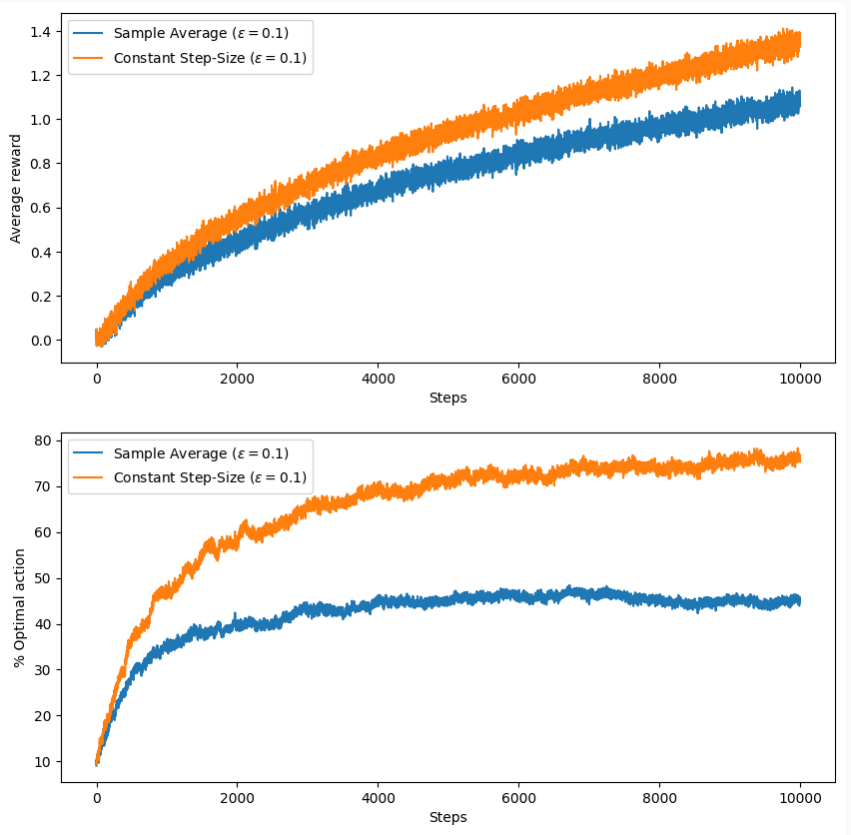
- $\rightarrow$ 2000 randomly generated 10-arm bandits
- $\rightarrow$ $d_k \sim \mathcal{N}(0, 1)$
- $\rightarrow$ $q_{*,t}(a) \sim \mathcal{N}(\sum_{k=1}^{t} d_k, 0)$
- $\rightarrow$ $R(a) \sim (q_*(a), 1)$



Constant Step-size Estimator

Experiment

- → non-stationary 10-arm agent
- → constant step-size value estimator w/ $\alpha = 0.1$
- → ϵ-greedy agent
- → % optimal action averaged over 2000 bandits



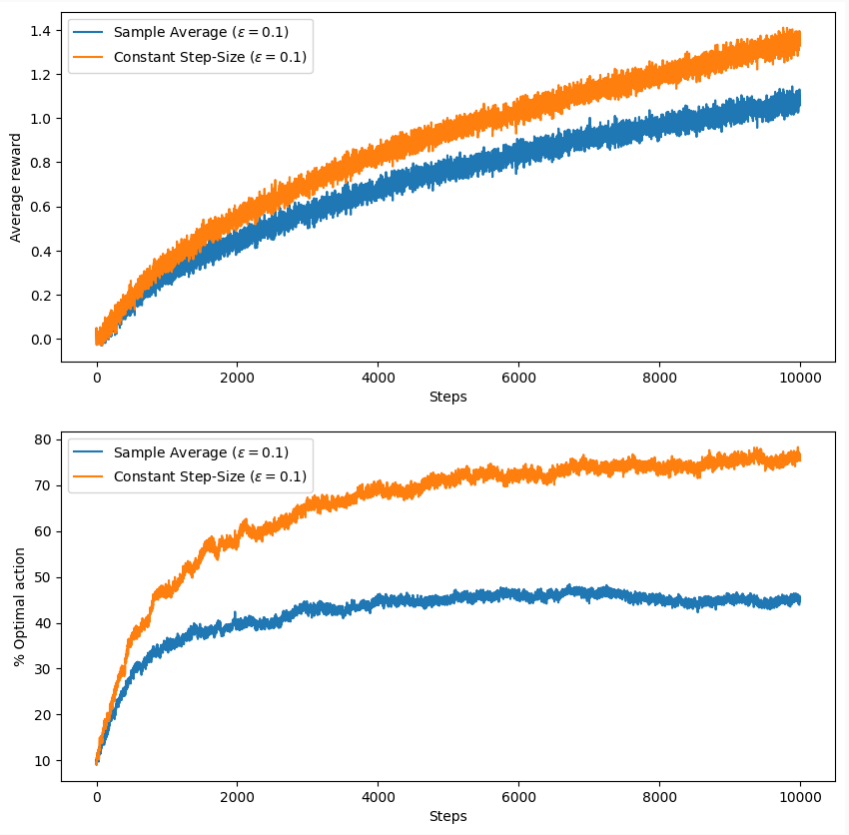
Constant Step-size Estimator

Observations

→ Sample Average is not an effective estimator for a non-stationary problem because it places equal weight on old rewards which may no longer reflect current action values.

→ The constant step-size estimator is much better at tracking correct action values because it exponentially discounts old rewards.

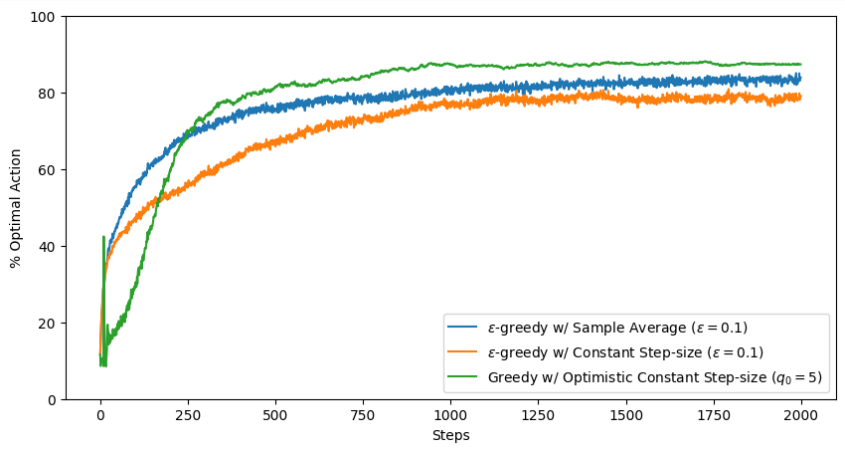
→ Note that this approach is not efficient in the stationary case.



Constant Step-size Estimator and Optimistic Initial Values

Experiment

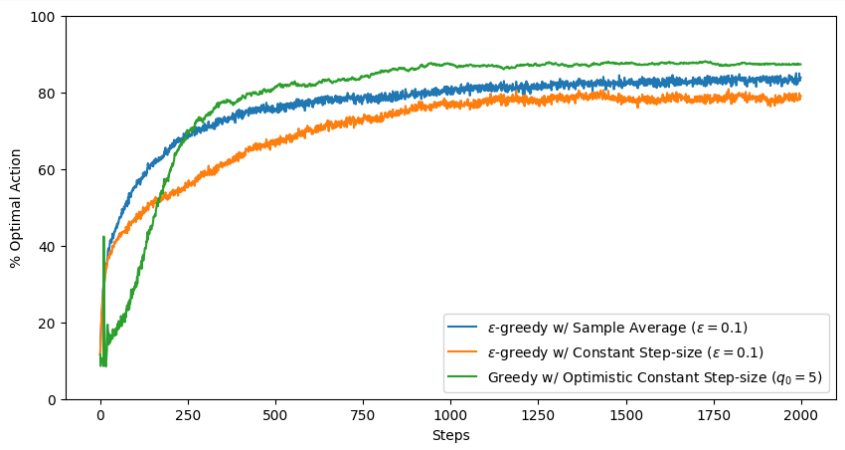
- $\rightarrow$ stationary 10-arm agent
- $\rightarrow$ constant step-size value estimator w/ $\alpha = 0.1$
- $\rightarrow$ ϵ-greedy agent
- $\rightarrow$ % optimal action averaged over 2000 bandits



Constant Step-size Estimator and Optimistic Initial Values

Observations

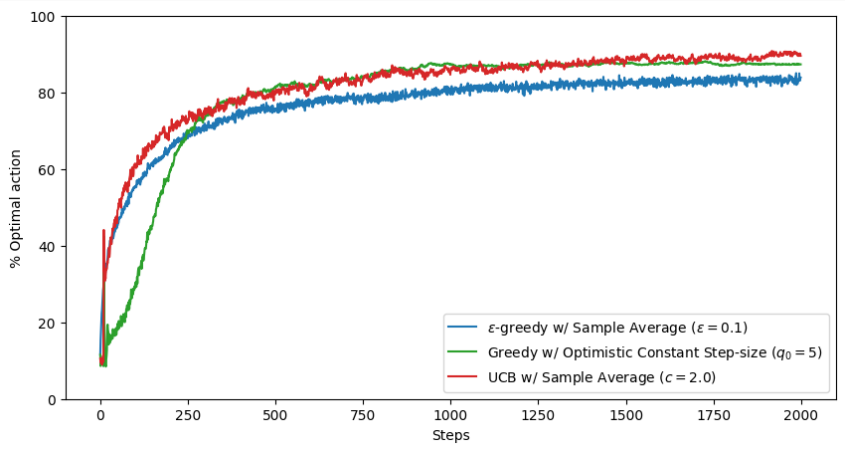
- → For the stationary problem, constant step-size is less efficient than sample average.
- → Using optimistic initial values with constant step-size estimates encourages a lot of early exploration and helps overcome the initial inefficiency, eventually performing better than the sample average estimates.
- → Note that using optimistic values with $\epsilon = 0$ would be counterproductive, as the two mechanisms would work against each other in the early steps.



Upper-Confidence-Bound Action Selection (UCB)

Experiment

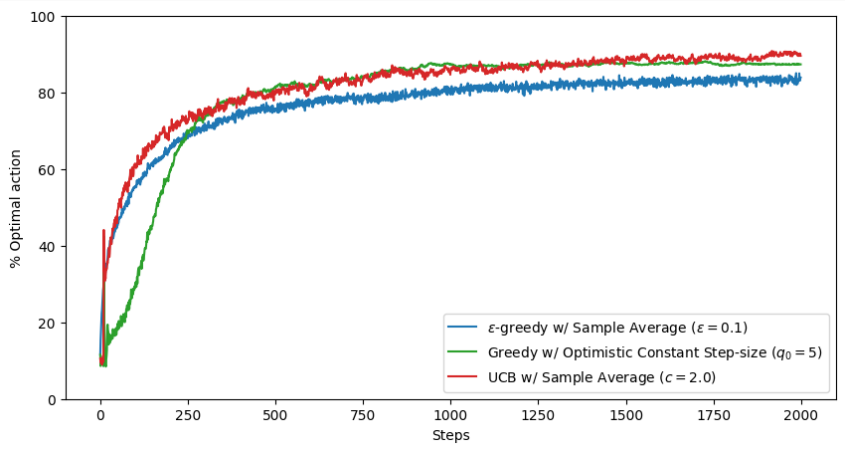
- → stationary 10-arm agent
- → sample average estimator
- → UCB agent
- → reward averaged over 2000 bandits



Upper-Confidence-Bound Action Selection (UCB)

Observations

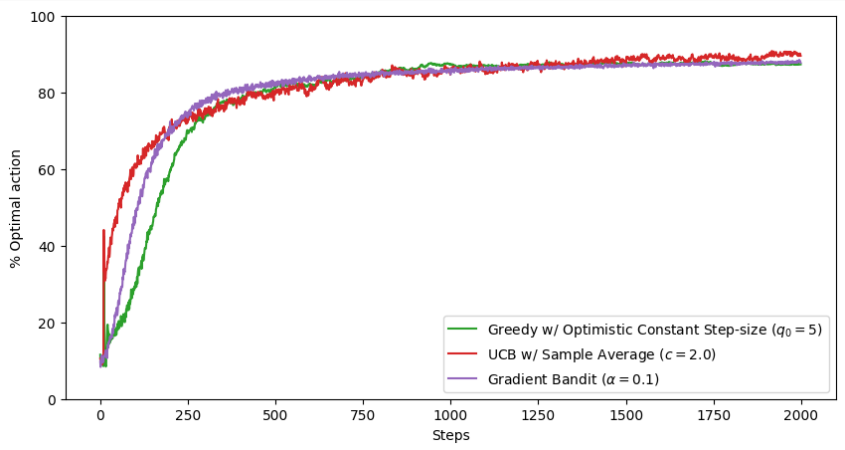
- → UCB performs better than ϵ-greedy, using the sample average estimator.
- → UCB also outperforms the greedy constant step-size with optimistic initial values.
- → Does not perform very well on non-stationary problems, since the $1/\sqrt{n}$ does not accurately reflect estimate uncertainty in the non-stationary case.



Gradient Bandit

Experiment

- $\rightarrow$ stationary 10-arm agent w/ $q_*(a) = 4$
- $\rightarrow$ Gradient Bandit w/ $\alpha = 0.1$
- $\rightarrow$ reward averaged over 2000 bandits



Summary

Environments

1. Stationary k-armed Bandit
2. Non-Stationary k-armed Bandit

Estimators

1. Sample-Average
2. Constant Step-Size
3. Constant Step-Size with Optimistic Initial Values

Agents

1. Greedy
2. ϵ-Greedy
3. UCB
4. Gradient Bandit

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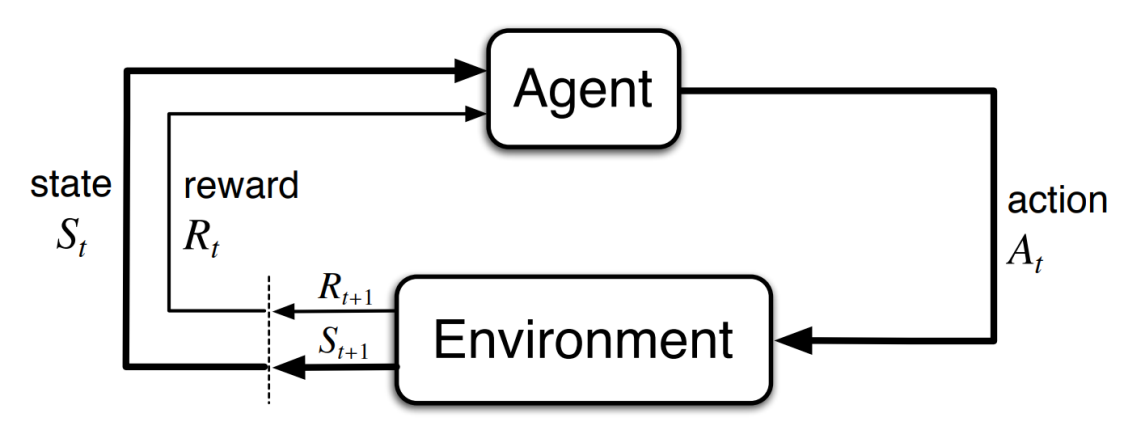
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Chapter 3
Finite Markov Decision Processes

Agent-Environment Interface



agent learner and decision maker

environment everything outside the agent

$S_t \in \mathcal{S}$ state of the environment at time t

$A_t \in \mathcal{A}(s)$ action at time t

$R_t \in \mathcal{R} \subset \mathbb{R}$ reward received as a consequence of action A_{t-1}

Markov Decision Processes (MDP)

Markov Property

The probability of each possible value for S_t and R_t is completely determined by the preceding state and action S_{t-1} and A_{t-1}.

Dynamics Function

The function $p : \mathcal{S} \times \mathcal{R} \times \mathcal{S} \times \mathcal{A} \mapsto [0, 1]$ is defined as

$$p(s', r \mid s, a) \doteq Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\}$$

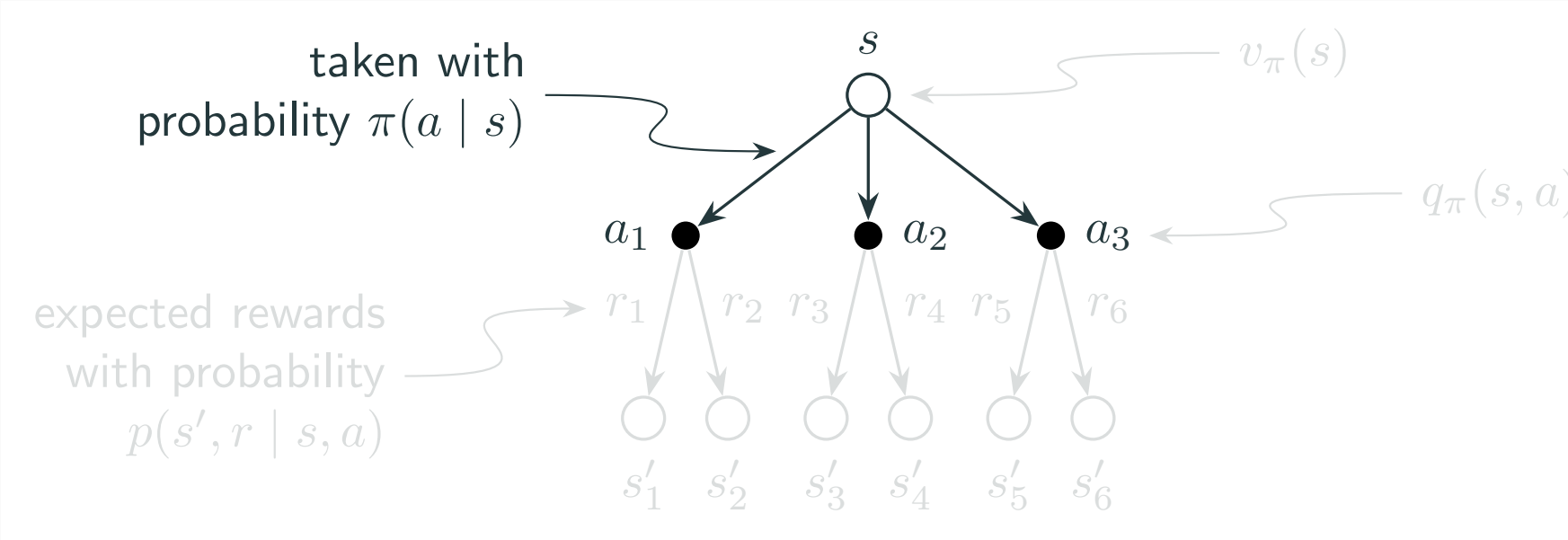
and completely defines the dynamics of the MDP.

Policies and Value Functions

Policy

Mapping from states to probability of selecting each possible action.

$$\pi : \mathcal{S} \mapsto \{\text{probabilities over } \mathcal{A}(s)\}$$

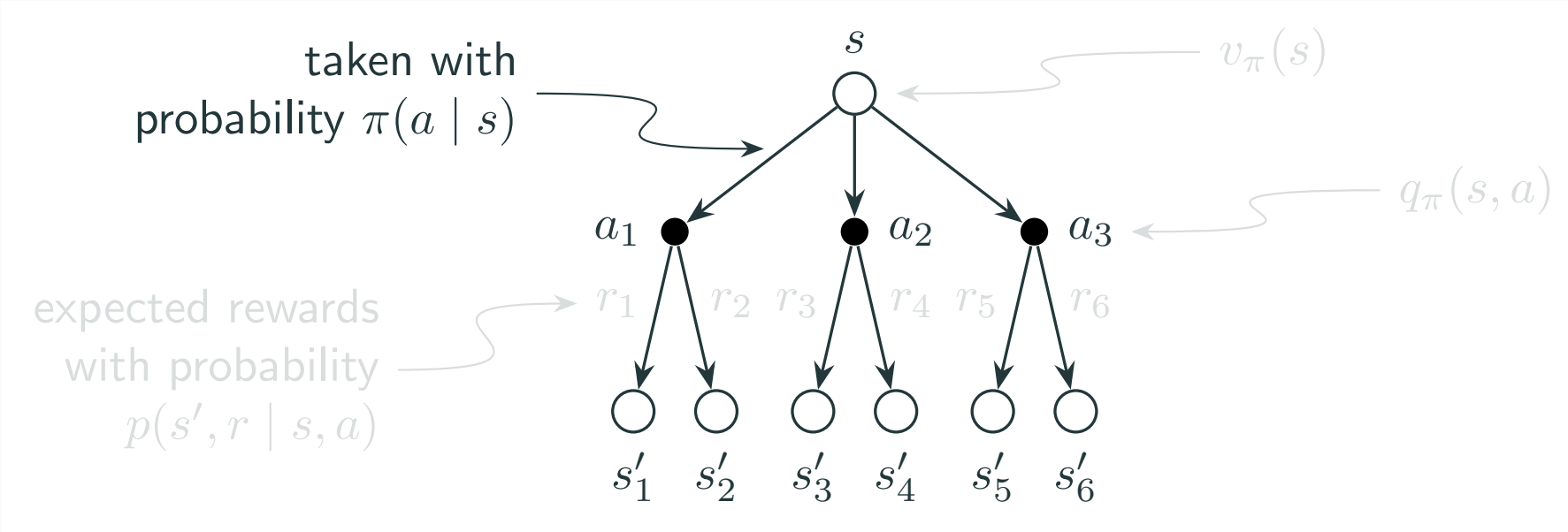


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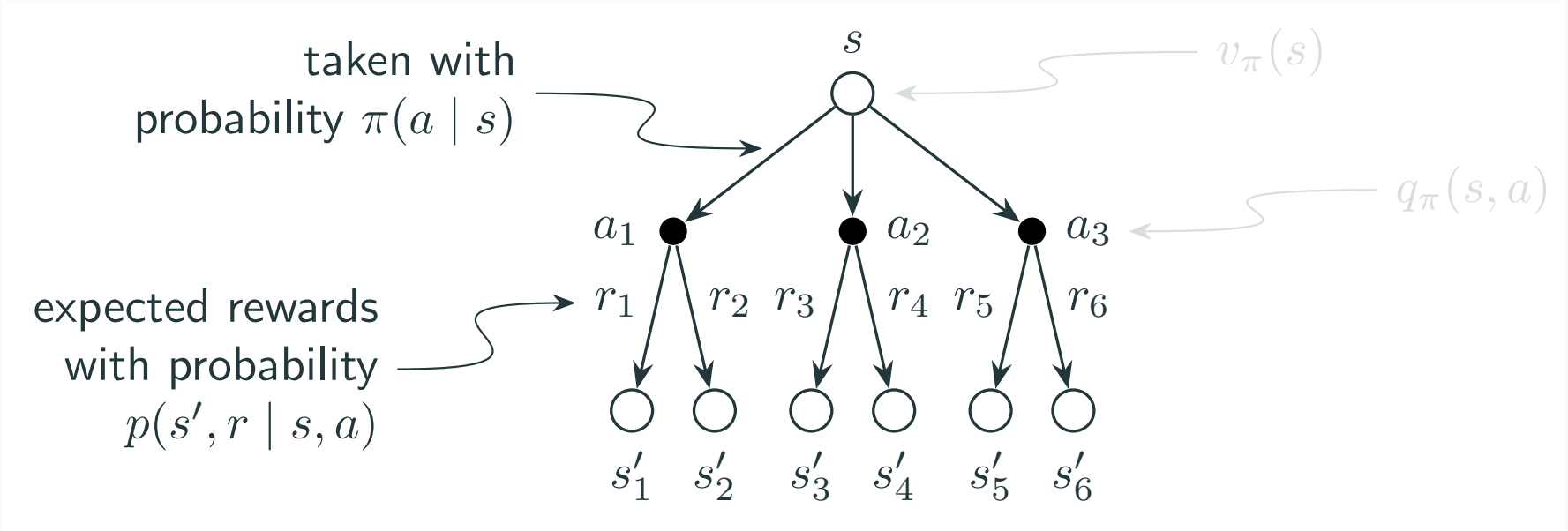


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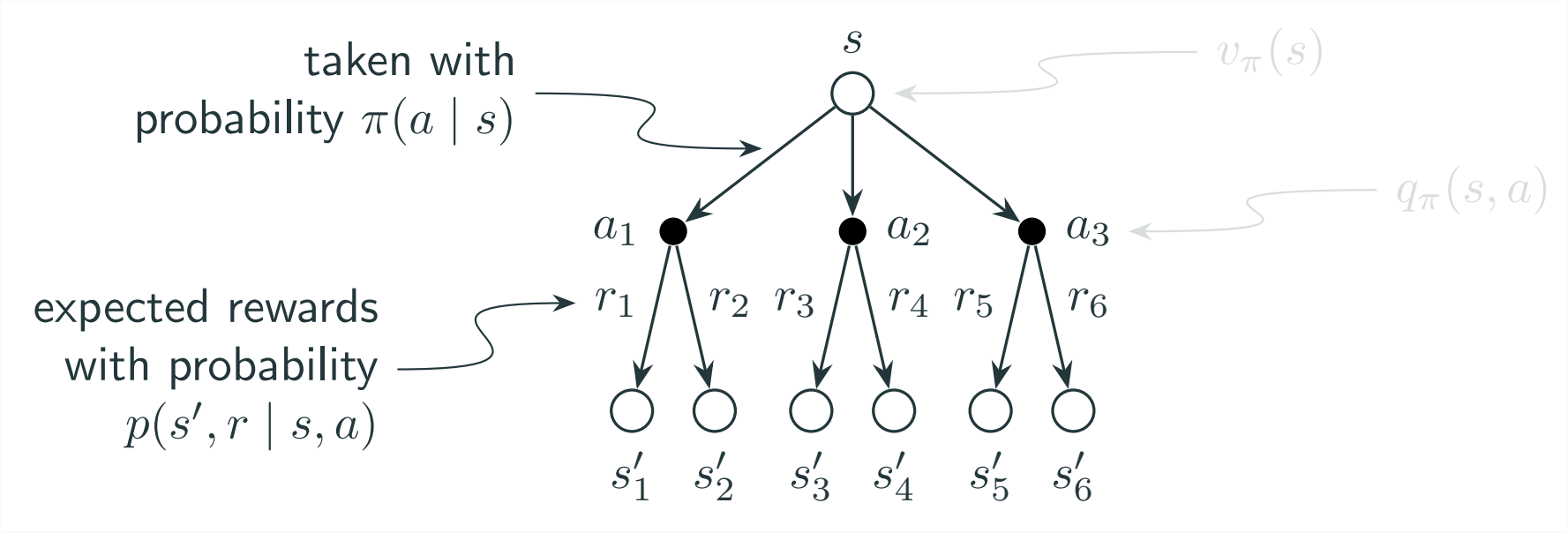
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State value function

Expected return when starting in a given state s and following policy π thereafter.

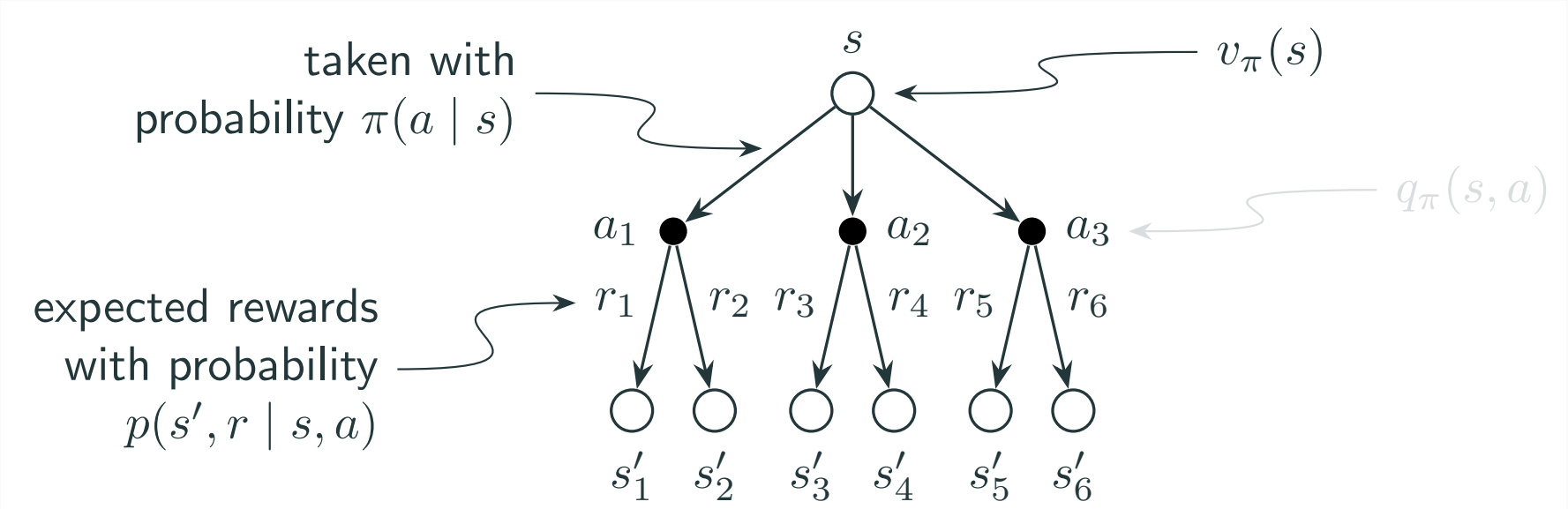
$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$$



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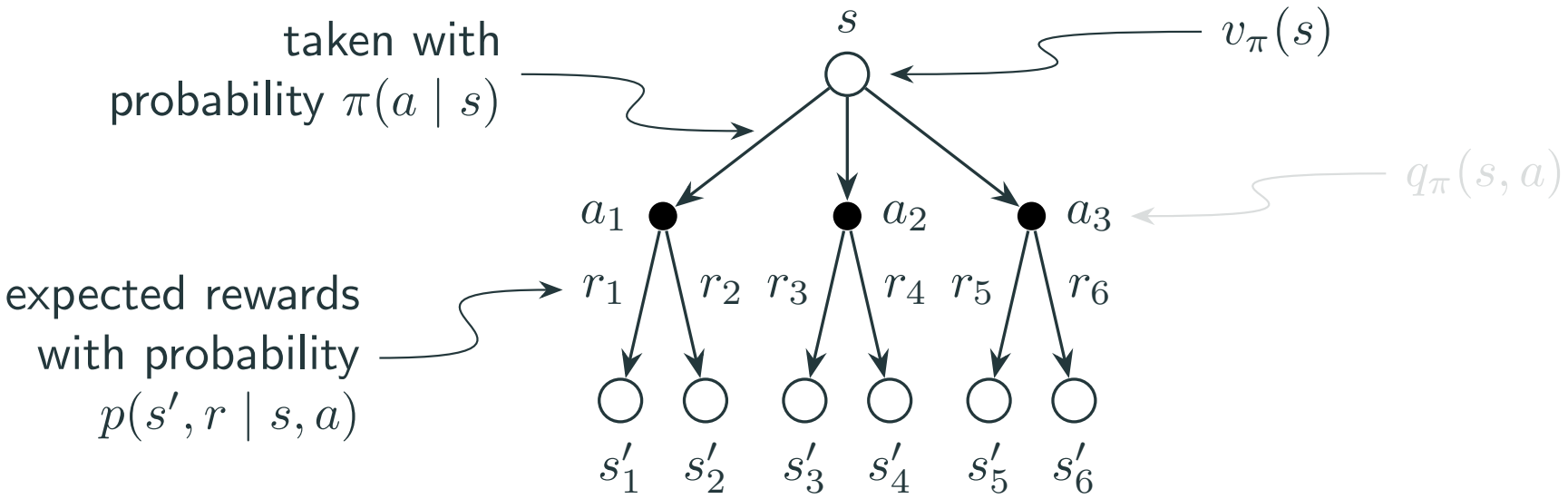
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Action value function

Expected return when starting in a given state s, taking an action a, then following policy π thereafter.

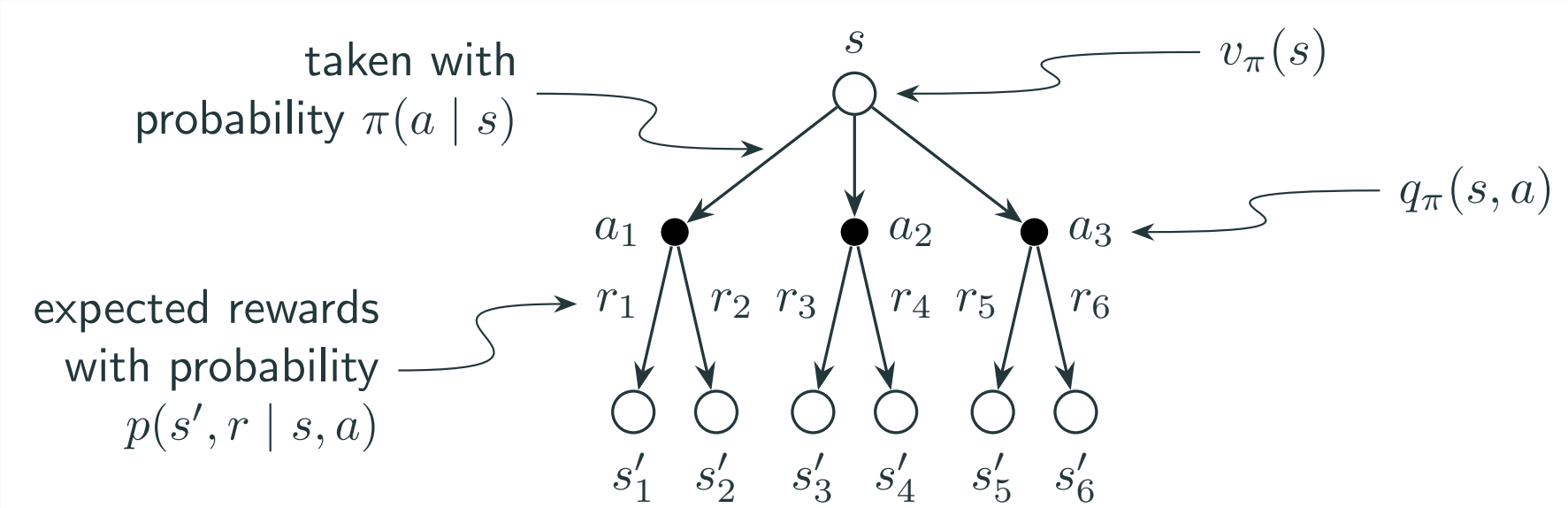
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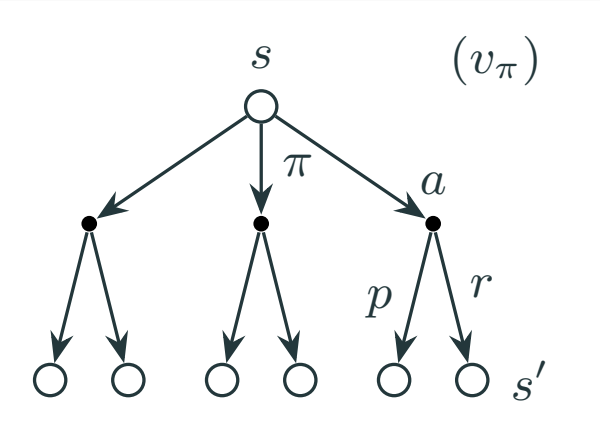
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Bellman Equations

$$v_\pi(s) = \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma v_\pi(s') \right]$$

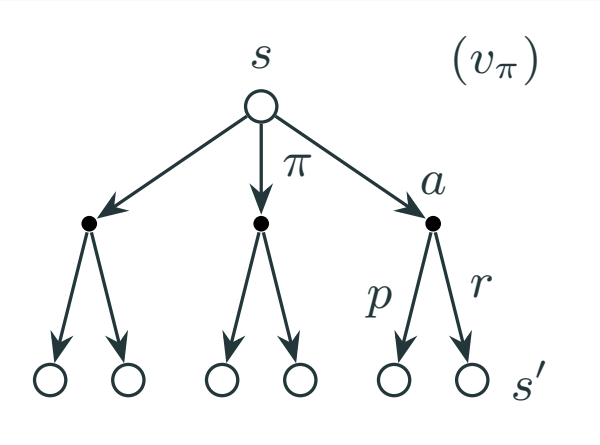
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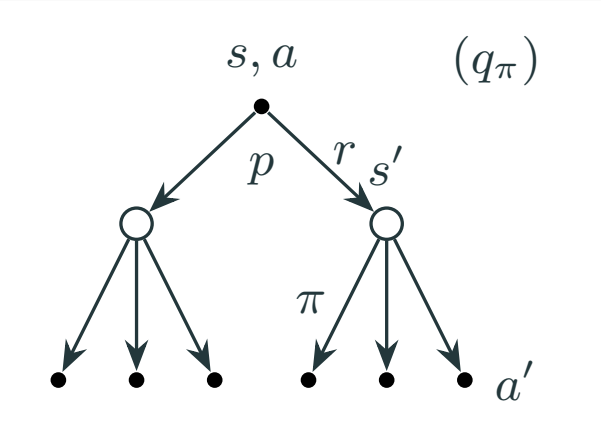


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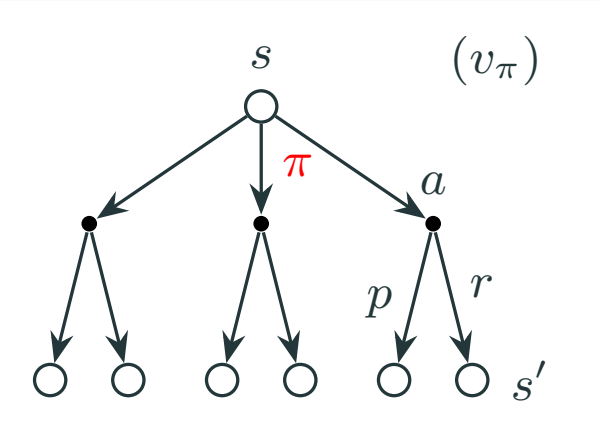




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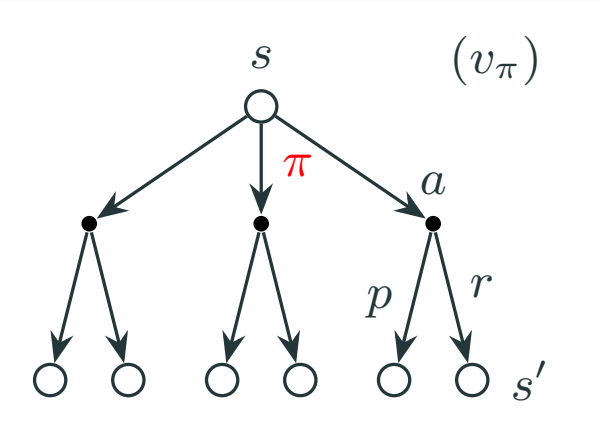
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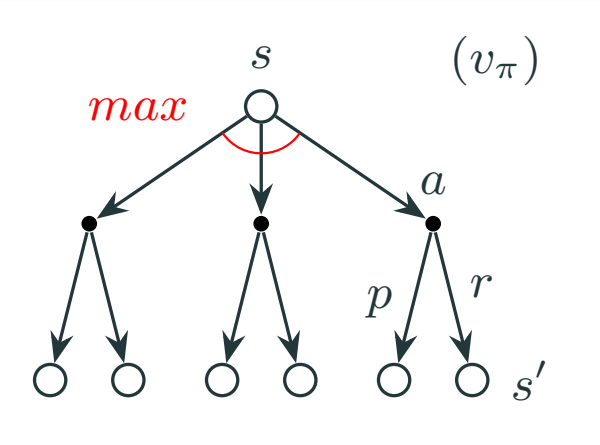
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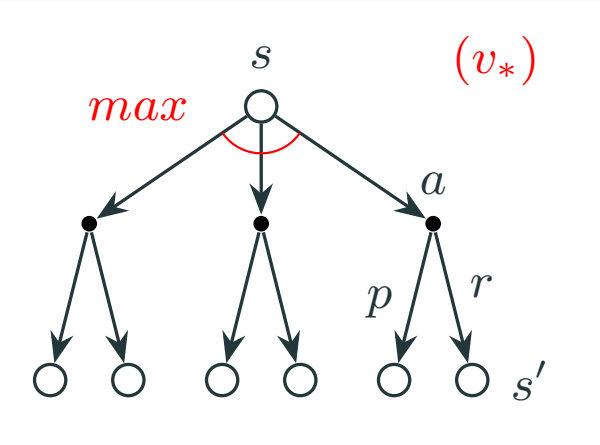
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Optimal Policies and Optimal Value Functions

Optimal Policy

π_* is said to be optimal if its expected return is greater than or equal to that of all other policies for all states:

$$v_{\pi_*}(s) \geq v_{\pi'}(s) \qquad \forall s \in \mathcal{S}, \forall \text{ policies } \pi'$$

Optimal Value Functions

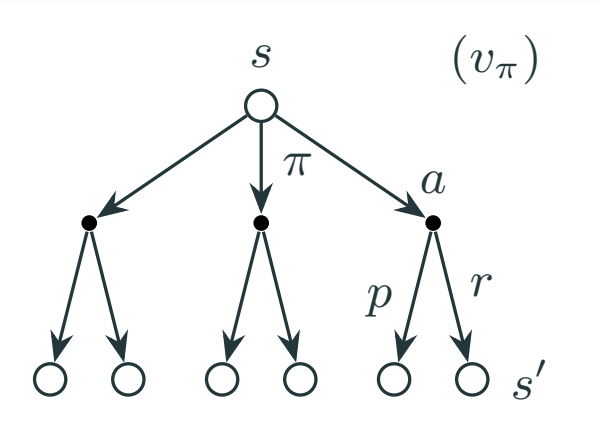
Value functions associated with optimal policies:

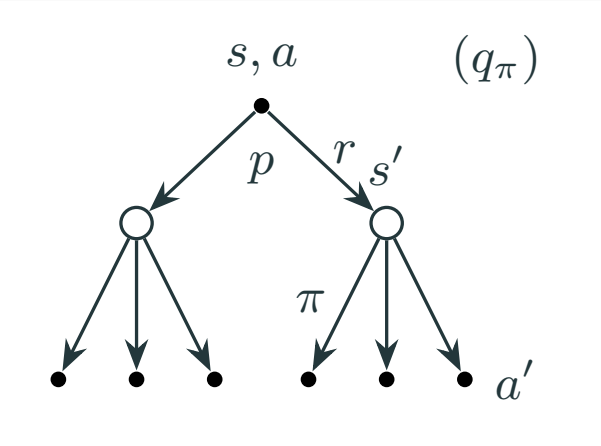
$$\begin{aligned} v_*(s) = v_{\pi_*}(s) &= \max_\pi v_\pi(s) && \forall s \in \mathcal{S} \\ q_*(s,a) = q_{\pi_*}(s,a) &= \max_\pi q_\pi(s,a) && \forall a \in \mathcal{A}(s), \forall s \in \mathcal{S} \end{aligned}$$

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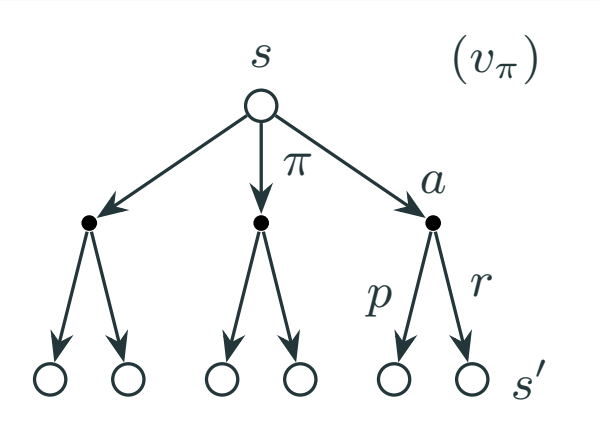


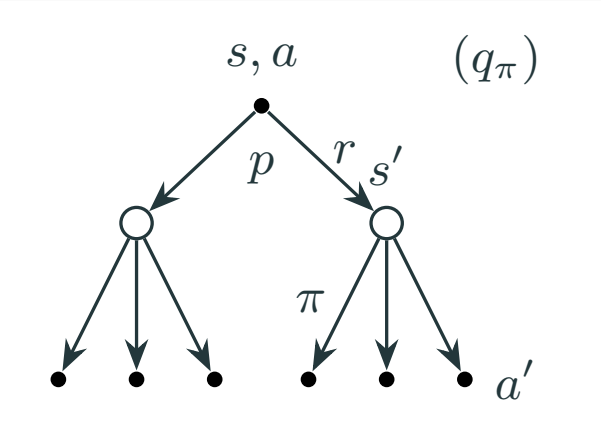


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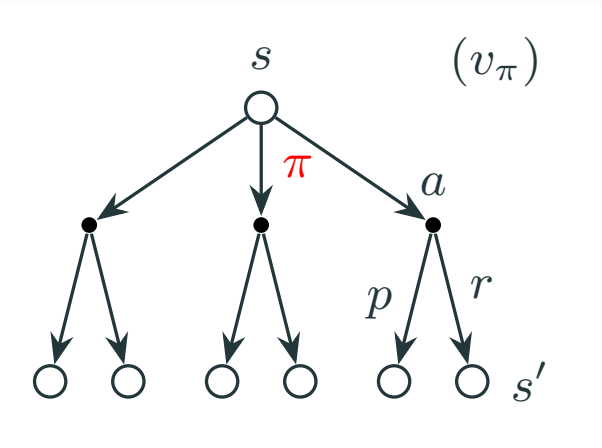


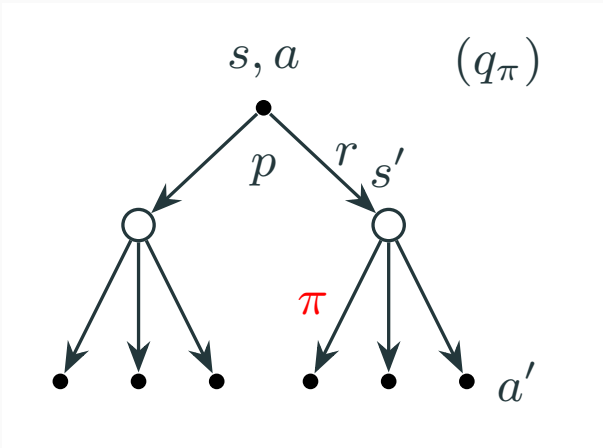
Optimal Policies and Optimal Value Functions

Bellman Equations

$$v_\pi(s) = \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma v_\pi(s') \right]$$

$$q_\pi(s, a) = \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma \sum_{a' \in \mathcal{A}(s')} \pi(a' \mid s') q_\pi(s', a') \right]$$





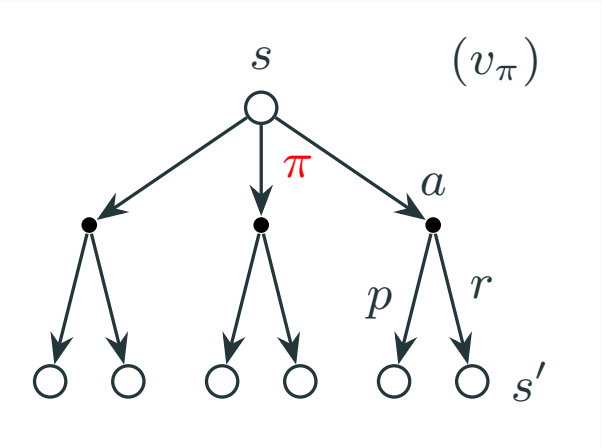
Bellman Equations

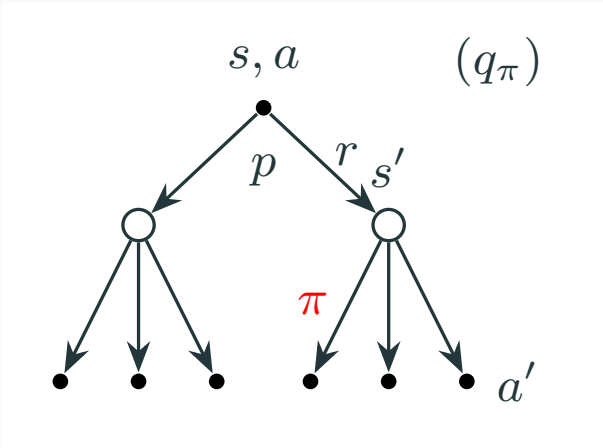
$$v_\pi(s) = \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma v_\pi(s') \right]$$

$$\max_a$$

$$q_\pi(s, a) = \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma \sum_{a' \in \mathcal{A}(s')} \pi(a' \mid s') q_\pi(s', a') \right]$$

$$\max_a$$





Optimal Policies and Optimal Value Functions

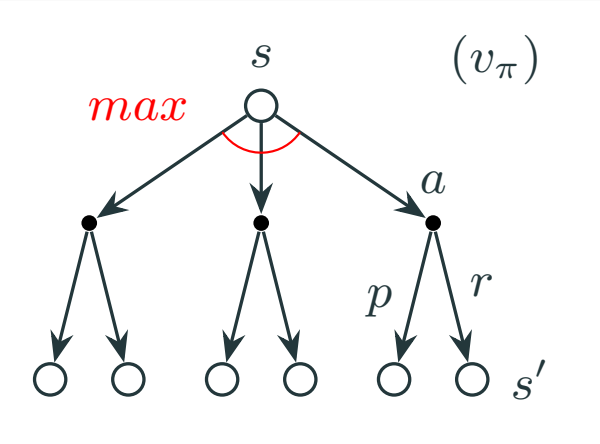
Bellman Equations

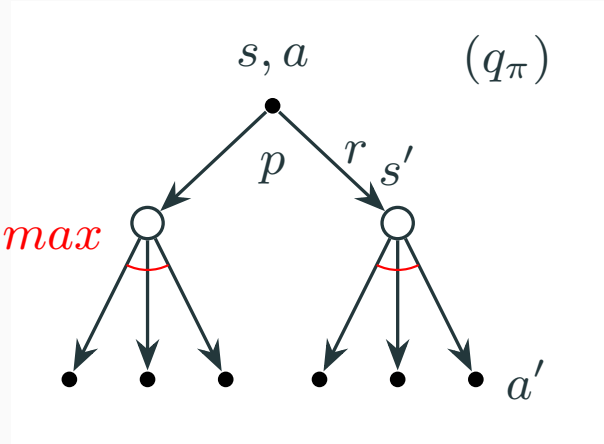
$$v_\pi(s) = \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma v_\pi(s') \right]$$

$$\max_a$$

$$q_\pi(s, a) = \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma \sum_{a' \in \mathcal{A}(s')} \pi(a' \mid s') q_\pi(s', a') \right]$$

$$\max_a$$





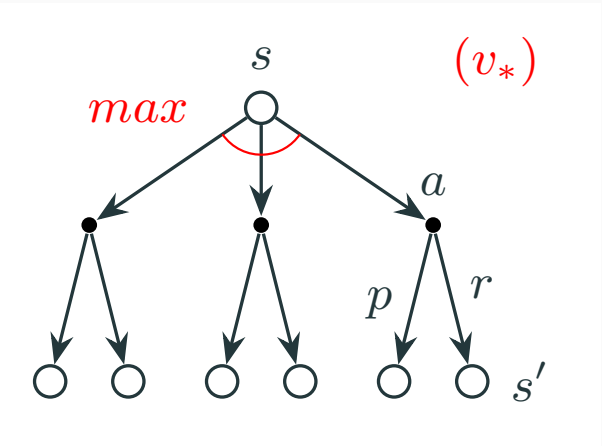
Bellman Equations

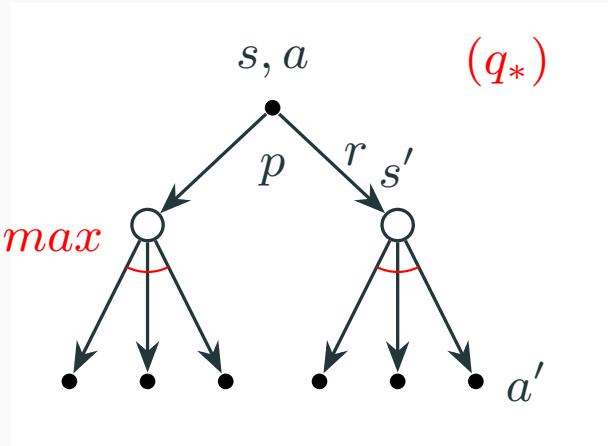
$$v_\pi(s) = \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma v_\pi(s') \right]$$

$\max_a$

$$q_\pi(s, a) = \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma \sum_{a' \in \mathcal{A}(s')} \pi(a' \mid s') q_\pi(s', a') \right]$$

$\max_a$

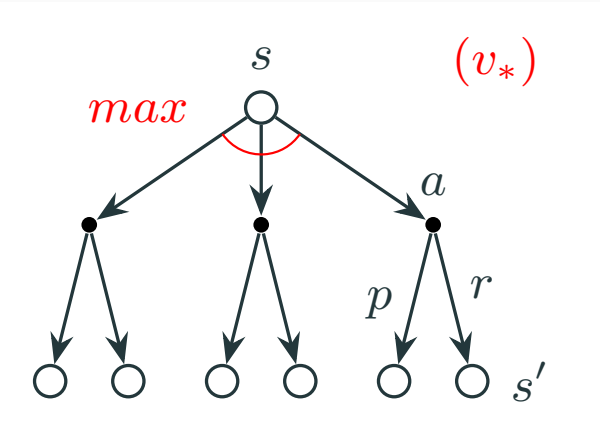


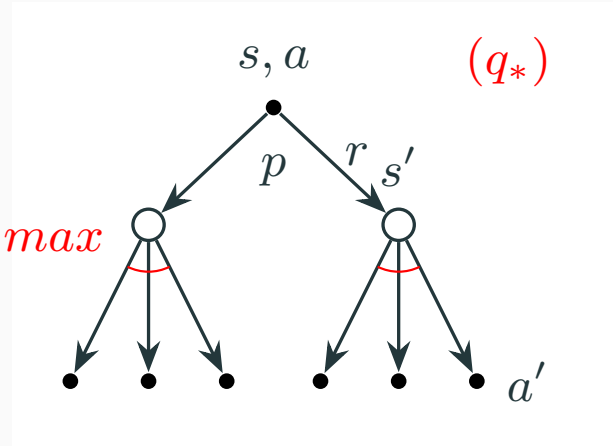


Bellman Optimality Equations

$$v_*(s) = \max_{a \in \mathcal{A}(s)} \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma v_*(s') \right]$$

$$q_*(s, a) = \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r \mid s, a) \left[r + \gamma \max_{a' \in \mathcal{A}(s')} q_*(s', a') \right]$$





Chapter 4
Dynamic Programming